

Lab: Case Study Analysis — Neural Bases of Social Communication

Objective

Analyze experimental and clinical cases illustrating the neural systems supporting communication, applying Active Inference to interpret each.

Part 1: Speaker-Listener Neural Coupling

Case: In a hyperscanning fMRI study, a speaker narrates a personal story while a listener hears the recording. The listener's brain activity in auditory, language, and prefrontal regions mirrors the speaker's with a 1-3 second lag. The degree of coupling predicts the listener's subsequent comprehension accuracy.

- How does Active Inference explain this coupling as generalized synchrony between two generative models?
- Why does a temporal lag exist, and what does it represent cognitively?
- Predict what would happen to the coupling if the story were told in a language the listener does not understand.

Write your response here

Part 2: Mirror Neurons and Action Understanding

Case: TMS (transcranial magnetic stimulation) is applied to a participant's inferior frontal gyrus (mirror neuron region) while they observe a hand grasping an object. Normally, observing this action produces a measurable increase in motor-evoked potentials (MEPs) in

the observer's hand muscles. After TMS disruption, this motor facilitation is abolished, and the participant shows impaired ability to predict the goal of ambiguous grasping actions.

- Interpret the motor facilitation as motor simulation — the observer's generative model predicting the proprioceptive consequences of the observed action.
- How does abolishing this facilitation impair goal prediction?
- Does this evidence prove mirror neurons are necessary for action understanding, or could alternative mechanisms compensate?

Write your response here

Part 3: False Belief Task and Theory of Mind

Case: A child watches a puppet show: Sally puts a marble in a basket and leaves. Anne moves the marble to a box. Sally returns. When asked "Where will Sally look for the marble?", children under 4 typically answer "in the box" (reality), while children over 4 answer "in the basket" (Sally's false belief).

- How does Active Inference model the false belief task? What hierarchical level of the generative model must be engaged?
- What changes in the child's brain around age 4 that enables false belief reasoning? (Consider TPJ maturation, precision on higher-order social models.)
- How does this task relate to "recursive" inference — modeling another agent's model of the world?

Write your response here

Part 4: Social Communication in Autism

Case: Adult with autism spectrum condition participates in a conversation. Eye-tracking shows reduced fixation on the speaker's eyes and increased fixation on the mouth. The participant reports understanding the literal content perfectly but missing "the point" of ironic or sarcastic remarks. fMRI shows typical activation in language regions but reduced activation in the mentalizing network (TPJ, mPFC).

- How does the Active Inference framework explain the pattern of preserved literal but impaired pragmatic comprehension?
- How might reduced mentalizing network activation reflect lower precision on social-inferential models?
- Is the autism profile best described as a deficit or a different optimization profile? Defend your answer.

Write your response here

Part 5: Synthesis

Write a 200-word synthesis explaining how these four cases collectively demonstrate that communication requires coupled inference at multiple levels — from motor simulation to higher-order belief modeling.

Write your response here

Lab Summary

Part	Skill Developed	Case Type
1	Neuroimaging interpretation	Speaker-listener neural coupling
2	Experimental reasoning	Mirror neurons and TMS disruption
3	Developmental analysis	Theory of Mind and false belief
4	Clinical analysis	Autism and pragmatic communication
5	Integrative synthesis	Multi-level social inference